

NOETICA MASTER BLUEPRINT

PART 02

Knowledge Universe & Self-Evolving Taxonomy

Purpose

The purpose of this component is to provide Noetica with a continuously evolving representation of human knowledge.

Unlike traditional classification systems, Noetica does not assume that fields, disciplines, or categories are fixed.

Knowledge evolves.

The taxonomy must evolve with it.

Core Principle

Traditional systems organize knowledge as folders.

Example:

Science |—— Biology |—— Chemistry |—— Physics

This structure is simple but fundamentally flawed.

Knowledge does not behave like folders.

Knowledge behaves like networks.

Noetica Principle

Knowledge is a graph.

Not a hierarchy.

Not a tree.

Not a folder structure.

Ideas can belong to multiple fields simultaneously.

A discovery may influence dozens of disciplines.

A field may emerge from the interaction of previously unrelated domains.

Objectives

The Knowledge Universe must:

1. Represent all domains of human knowledge.
 2. Continuously discover new disciplines.
 3. Detect relationships between fields.
 4. Detect field convergence.
 5. Detect field divergence.
 6. Detect emerging topics.
 7. Detect declining topics.
 8. Support forecasting.
 9. Support discovery ranking.
 10. Support cross-disciplinary intelligence.
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The Knowledge Universe

Noetica attempts to represent the entire known knowledge ecosystem.

This includes both historical and emerging knowledge domains.

Top-Level Domains

These domains are organizational conveniences.

They are not permanent truths.

Formal Sciences

Focus:

Abstract systems and reasoning.

Includes:

- Mathematics
 - Statistics
 - Logic
 - Information Theory
 - Systems Theory
 - Complexity Science
 - Decision Theory
 - Game Theory
 - Computational Theory
 - Network Science
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Physical Sciences

Focus:

Physical reality.

Includes:

- Physics
 - Chemistry
 - Materials Science
 - Earth Science
 - Atmospheric Science
 - Oceanography
 - Planetary Science
 - Astronomy
 - Astrophysics
 - Cosmology
 - Energy Science
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Life Sciences

Focus:

Living systems.

Includes:

- Biology
 - Genetics
 - Genomics
 - Transcriptomics
 - Proteomics
 - Metabolomics
 - Neuroscience
 - Systems Biology
 - Synthetic Biology
 - Evolutionary Biology
 - Ecology
 - Biotechnology
 - Agriculture
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Medical Sciences

Focus:

Human health and medicine.

Includes:

- Medicine
 - Oncology
 - Neurology
 - Immunology
 - Infectious Disease
 - Precision Medicine
 - Public Health
 - Regenerative Medicine
 - Longevity Research
 - Clinical Research
 - Drug Discovery
 - Epidemiology
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Computational Sciences

Focus:

Computation and information processing.

Includes:

- Computer Science
- Artificial Intelligence

- Machine Learning
 - Deep Learning
 - Natural Language Processing
 - Robotics
 - Computer Vision
 - Cybersecurity
 - Data Science
 - Human Computer Interaction
 - Distributed Systems
 - Quantum Computing
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Engineering Sciences

Focus:

Applied systems.

Includes:

- Mechanical Engineering
 - Electrical Engineering
 - Civil Engineering
 - Chemical Engineering
 - Aerospace Engineering
 - Biomedical Engineering
 - Systems Engineering
 - Manufacturing
 - Nanotechnology
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Social Sciences

Focus:

Human behavior and society.

Includes:

- Economics
- Finance
- Psychology
- Sociology
- Anthropology
- Political Science
- Human Geography
- Education
- Demography
- Communication Studies

Humanities

Focus:

Human meaning and culture.

Includes:

- Philosophy
- History
- Literature
- Linguistics
- Religious Studies
- Archaeology
- Ethics
- Cultural Studies

Creative Disciplines

Focus:

Creative expression and design.

Includes:

- Design
- Architecture
- Music
- Film
- Visual Arts
- Performing Arts
- Game Design
- Creativity Research

Interdisciplinary Domains

The future of knowledge often emerges between disciplines.

Examples:

Computational Biology

Mathematics + Biology

Neuroeconomics

Neuroscience + Economics

Econophysics

Physics + Economics

AI Drug Discovery

AI + Biology + Chemistry

Quantum Biology

Quantum Physics + Biology

Digital Humanities

AI + History + Linguistics

Synthetic Life

Biology + Engineering + AI

Computational Social Science

Computer Science + Sociology

AI-Assisted Science

AI + Scientific Research

Knowledge Nodes

The fundamental unit of the Knowledge Universe.

Examples:

Calculus

CRISPR

Transformer Architecture

Bayesian Inference

General Relativity

Protein Folding

Synthetic Cells

Each becomes a node.

Knowledge Node Types

Field

Subfield

Topic

Discovery

Theory

Technology

Methodology

Dataset

Research Program

Organization

Researcher

Conference

Grant

Patent

Startup

Government Initiative

Knowledge Relationships

Nodes are connected through relationships.

Examples:

influences

depends_on

derived_from

enabled_by

contradicts

supports

accelerates

commercialized_by

funded_by

implemented_by

competes_with

Self-Evolving Taxonomy Engine

The taxonomy must continuously update itself.

Daily Questions

What fields are emerging?

What fields are merging?

What fields are fragmenting?

What fields are declining?

What concepts are becoming dominant?

What concepts are becoming obsolete?

Emergence Detection

The system continuously monitors:

Research Papers

Patents

Conferences

Open Source Projects

Funding Programs

Research Grants

Government Initiatives

Datasets

Scientific Roadmaps

Example

Before 2017:

Transformer Architectures

Status:

Tiny Topic

After 2017:

Transformer Architectures

↓

Large Research Area

↓

Foundation Models



AI Agents



AI Scientists

The taxonomy should update automatically.

Field Lifecycle

Fields evolve similarly to discoveries.

Stages:

Emerging

Growing

Expanding

Established

Dominant

Declining

Historical

Field Convergence Detection

Examples:

AI

+ Biology



AI Drug Discovery

Neuroscience

+ Economics



Neuroeconomics

Physics

+ Biology



Quantum Biology

The system must detect these automatically.

Field Divergence Detection

Examples:

Artificial Intelligence



Machine Learning



Deep Learning



Foundation Models



Agentic Systems



AI Scientists

One field can produce multiple descendants.

Knowledge Growth Metrics

Each field receives metrics.

Examples:

Publication Growth

Citation Growth

Funding Growth

Patent Growth

Researcher Growth

Startup Growth

Open Source Growth

Dataset Growth

Knowledge Importance Metrics

Field Significance

Cross-Disciplinary Influence

Civilizational Relevance

Research Momentum

Technological Impact

Scientific Impact

Educational Impact

Economic Impact

Future-Proofing Requirement

The taxonomy must never assume that the current structure of knowledge is final.

New fields must be discoverable without changing the codebase.

No field should be permanently hardcoded as a special case.

Long-Term Objective

Build a living representation of human knowledge that continuously reorganizes itself as humanity discovers, creates, combines, and transforms ideas.

The taxonomy should not merely describe knowledge.

It should evolve alongside knowledge itself.